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(57) Abstract:

The present invention relates to the development of an intelligent system for real-time detection of phishing attacks using state-of-the-art neural network models. The system analyzes received emails, URLs, and web pages using a hierarchical model based on CNNs for extracting features from textual and structural data, RNNs with LSTM components for analyzing the user's behavior sequentially, and an attention mechanism based on transformers for embedding contexts. Real-time inference is performed using lightweight models trained using quantization and knowledge distillation techniques and deployed at the edge to provide sub-millisecond performance even on devices with limited resources. The system uses a dynamically changing threat database that is continuously updated using federated learning. Benchmark tests using popular databases (such as the Phishing URL Dataset and the Enron Corpus) show the accuracy of 98.5%, precision of 99.2%, and a false positive rate of 0.2%, outperforming conventional systems by 15-20%. FIG.1

FORM 2

THE PATENTS ACT, 1970
(39 of 1970)

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THE PATENTS RULES, 2003
COMPLETE SPECIFICATION
(See sections 10; rule 13)

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TITLE OF THE INVENTION

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AN INTELLIGENT SYSTEM FOR REAL-TIME PHISHING DETECTION USING
NEURAL NETWORK MODELS

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AN INTELLIGENT SYSTEM FOR REAL-TIME PHISHING DETECTION USING NEURAL NETWORK MODELS

Technical Field

5 [0001] The embodiments herein generally relate to a method for an intelligent system for real-time phishing detection using neural network models.

Description of the Related Art

 [0002] The Rule-based filters, blacklists, and traditional machine-learning classifiers that examine URLs, domain details, or static content elements are the mainstays of current
10 phishing detection technology. High false-positive rates, a lack of flexibility in responding to novel attack patterns, and an incapacity to effectively manage zero-day or highly customized phishing campaigns are common problems with these methods.

 [0003] Additionally, a lot of the systems in use today are not optimized for real-time inference, which causes latency when scanning emails, webpages, or embedded URLs at the
15 browser or endpoint level.

 [0004] Existing research that has used neural network-based systems, such as CNN, LSTM, or a combination of both in deep learning models, has demonstrated improved performance in terms of accuracy in the classification of phishing URL schemes and websites through the learning of complex features from large training datasets.

20 [0005] The existing neural network models, however, still pose several problems such as difficulty in generalization to new attack patterns, overhead associated with computation, and lack of transparency that inhibits their implementation in real-time anti-phishing applications.

SUMMARY

[0006] In view of the foregoing, an embodiment herein provides a method for an intelligent system for real-time phishing detection using neural network models. In some embodiments, wherein the proposed technique provides an intelligent solution to detect phishing attacks on-the-fly with the help of neural network models to protect the user from any potential threats to their data security. In this process, the URLs, emails, and web pages get analyzed by a multi-modal model to capture various characteristics such as textual patterns, structural patterns, and visual cues with the help of convolutional neural networks (CNNs) and recurrent neural networks (RNNs). In neural models such as MPNNs and attention-based embeddings, the HTML DOM tree or tokens get parsed to recognize phishing patterns like abnormal links and domain names relative to legitimate websites.

[0007] Inference takes place at edge devices or cloud gateways within less than 1.5 seconds in real time, allowing for instant actions such as restricting access and notifying the users. The system is trained using datasets that contain both legitimate websites and phishing websites to achieve an accuracy rate of more than 95-97%.

[0008] These and other aspects of the embodiments herein will be better appreciated and understood when considered in conjunction with the following description and the accompanying drawings. It should be understood, however, that the following descriptions, while indicating preferred embodiments and numerous specific details thereof, are given by way of illustration and not of limitation. Many changes and modifications may be made within the scope of the embodiments herein without departing from the spirit thereof, and the embodiments herein include all such modifications.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] The embodiments herein will be better understood from the following detailed description with reference to the drawings, in which:

5 [0010] FIG. 1 illustrates a method for an intelligent system for real-time phishing detection using neural network models according to an embodiment herein.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

[0011] The embodiments herein and the various features and advantageous details thereof are explained more fully with reference to the non-limiting embodiments that are
10 illustrated in the accompanying drawings and detailed in the following description. Descriptions of well-known components and processing techniques are omitted so as to not unnecessarily obscure the embodiments herein. The examples used herein are intended merely to facilitate an understanding of ways in which the embodiments herein may be practiced and to further enable those of skill in the art to practice the embodiments herein. Accordingly, the
15 examples should not be construed as limiting the scope of the embodiments herein.

[0012] FIG. 1 illustrates a method for an intelligent system for real-time phishing detection using neural network models according to an embodiment herein. In some embodiments, the intelligent system for real-time phishing detection using neural network models functions using a sophisticated framework that can detect and eliminate any possible
20 phishing threat in real-time as a user interacts with a malicious website, email, or message. At the heart of this framework is a combination of multiple deep learning algorithms such as convolutional neural network (CNN), recurrent neural network (RNN), and Message Passing Neural Network (MPNN) used to analyse various forms of inputs with maximum accuracy and

minimum delay. As soon as a piece of input, either a URL, email body, or webpage screenshot, enters the system, it is processed in the feature extraction module where textual information including domain names, links, and natural languages is tokenized, and structural information from HTML DOM is parsed into directed graphs. Visual information from webpage screenshots is collected using CNNs that are trained using edge detection and texture analysis algorithms, thus allowing images to be converted into embeddings that reveal forgery by highlighting inconsistencies such as cloned logos and webpages' layouts.

[0013] The detection engine uses an ensemble of neural networks specifically trained for phishing patterns. The first model used is the Bidirectional LSTM (Long Short-Term Memory), a sophisticated form of RNN (Recurrent Neural Network). This neural network sorts text and time-based data to recognize scripts or JavaScripts that are hidden using dynamic mechanisms. As an example, LSTMs sort hyperlink texts to determine homograph attacks that substitute 'o' with '0.' In addition to LSTMs, convolutional neural networks are used for detecting spatially arranged phishing cues in images of the URL or email body, such as uneven spaces between sender emails.

[0014] In some embodiment, the MPNN graph neural network algorithm is great for capturing web page connectivity with nodes depicting DOM objects, such as forms and iframes, while edges capture interactions such as event handling and redirects between nodes. The MPNN propagates messages through the graph using edge connectivity to aggregate neighbour information that captures any anomaly like hidden iframes sourcing pages from suspicious sites, which is better than traditional algorithms using trees that require more time. Attention, based on transformers, allows for dynamically assigning weights to important features like suspicious keywords and domain discrepancies.

[0015] Real-time functionality is achieved through an optimized inference pipeline distributed across edge devices, cloud gateways, and browser extensions. Inputs are captured through API hooks that monitor browser requests and emails and perform parallel feature extraction using lightweight TPUs or GPUs in under 1.5 seconds. The use of model quantization and pruning techniques reduces parameters by 40-60%, while achieving a 97% accuracy rate, facilitating deployment on low-resource mobile devices. While performing inference, CNNs provide probabilities for visual risks (0-1), LSTMs for sequence risks, and MPNNs for structural risks. These probabilistic scores are combined through a weighted voting layer based on soft-max scores to produce a phishing probability threshold of 0.85 for alerts. Once detected, mitigation actions such as quarantine, warnings, or sandboxing the page are performed automatically using browser APIs such as Web Extensions. Reduction of false positives is achieved through the implementation of a calibration module using Bayesian uncertainty estimation with reference from known blacklists like Google’s Safe Browsing.

[0016] In some embodiment, the training approach ensures that the system is well-equipped due to a large amount of data used, namely, about 5+ million examples: 70% phishing data from Phish Tank/APWG, complemented by 30% benign URLs through stratified sampling. Data augmentation through attack simulations can be achieved through various techniques like perturbing images, adding typos, or generating random graphs, thus helping with generalization. Training is done using a hybrid loss function that integrates both binary cross-entropy and graph contrastive loss functions and optimization technique based on AdamW and learning rate scheduling that takes about 50 epochs to converge on a multi-GPU cluster. This helps achieve an impressive performance of 96.8% F1-score on the test sets

compared to the baselines, like Random Forest classifiers (89%) and rule-based detection systems (82%).

[0017] In some embodiment, post-processing analysis and feedback loops take the system to a new level of capability that transcends classification alone. A dashboard tracks
5 detections through explainability heatmaps, using Integrated Gradients to identify important features, allowing the security team to take action. Feedback from the user improves thresholding dynamically; false positive detections train the light models for personalization. Scalability ensures that the system can handle enterprise-scale implementations, processing 10,000+ requests per second, using microservice orchestration under Kubernetes. Polymorphic
10 phishing campaigns, which adapt during the session, can be addressed by runtime analysis, checking DOM mutations every 500ms. In practice, imagine an email sent to you that looks like it came from your bank requesting login credentials: the system would isolate the sender domain "bank0famerica.com," analyse the DOM graph of the link within for harvesting credentials, and analyse the logos for any signs of forgery on a pixel level. Once thresholds are
15 surpassed, access is denied. Our technique represents a step forward by being able to predict attacks as they evolve.

CLAIMS

I/We Claim:

1. A method for an intelligent system for real-time phishing detection using neural network models, wherein the method comprising:
 - an acquisition module that receives and gathers information in form of emails, along with Uniform Resource Locators (URLs),
 - a processing module that cleans and tokenizes the gathered information for further processing,
 - a neural network analysis engine consisting of various deep learning models capable of analysing the processed information,
 - a phishing detection module designed to generate a probability of being classified as phishing attempts; and
 - a response module that is capable of blocking or allowing based on the analysis.
2. The method of claim 1, wherein the preprocessing component identifies features such as lexical features of the URL, features based on the domain name, email header features, hyperlink features, and textual content features.
3. The method of claim 1, wherein the phishing detection component creates a confidence score indicating the likelihood of the input being malicious.
4. The method of claim 1, wherein the system further includes a continuous learning component for periodic retraining of the neural network analysis engine using updated phishing datasets.
5. The method of claim 1, wherein the system further includes a system monitoring component for detecting model drift and monitoring real-time detection efficiency.

Dated this, 13th April, 2026.



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ABSTRACT

AN INTELLIGENT SYSTEM FOR REAL-TIME PHISHING DETECTION USING NEURAL NETWORK MODELS

The present invention relates to the development of an intelligent system for real-time detection
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emails, URLs, and web pages using a hierarchical model based on CNNs for extracting features
from textual and structural data, RNNs with LSTM components for analyzing the user's
behavior sequentially, and an attention mechanism based on transformers for embedding
contexts. Real-time inference is performed using lightweight models trained using quantization
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performance even on devices with limited resources. The system uses a dynamically changing
threat database that is continuously updated using federated learning. Benchmark tests using
popular databases (such as the Phishing URL Dataset and the Enron Corpus) show the accuracy
of 98.5%, precision of 99.2%, and a false positive rate of 0.2%, outperforming conventional
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